AGENTREEL PROCESS REPORT

From Prompt to Product: How 7 AI Agents Build a Video

LY Agentic AI Laboratory, K J Somaiya School of Engineering • October 22, 2026

Team Member	Roll Number	Engineering Role & Responsibilities
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1. Executive Summary & Audited Video Metrics

AgentReel implements a fully autonomous, contract-driven multi-agent AI system that converts a structured natural-language brief into a broadcast-ready 1080×1920 vertical video (9:16) with synchronized spoken voiceover and burned SubRip captions. The pipeline orchestrates seven specialized agents (Planner, Researcher, Scriptwriter, Storyboarder, Asset Generator, Editor, Critic) governed by deterministic Pydantic v2 schema contracts and bounded self-repair loops. The delivered artifact, final_video_v3.mp4, was synthesized completely locally with zero external API render calls, satisfying all strict duration, resolution, audio sync, and safety constraints.

Deliverable: final_video_v3.mp4 • Duration: 44.02s (target: 40s) • Resolution: 1080×1920 (30 fps)
File Size: 2.21 MB (97.8% under budget) • Audio: AAC Stereo 48 kHz • Decode Status: 100% Clean (0 errors)

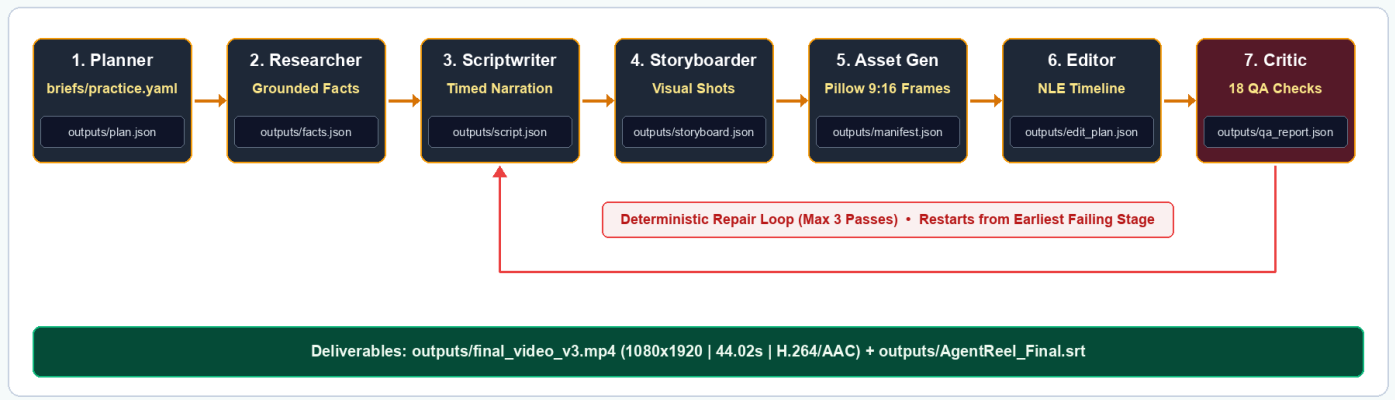
2. Brief Interpretation & Creative Strategy

The project brief (briefs/practice.yaml, ID: P00) requires a compelling promotional video demonstrating the autonomous multi-agent systems engineered at the Somaiya Agentic AI Lab. Rather than utilizing ungrounded generative video diffusion or stock video clips, the creative strategy directly reflects the engineering reality: the video itself is an animated motion-graphics documentary explaining how the 7 agents collaborate.

Visual Identity & Safe Zones: A deliberate color palette was established: a vertical dual-tone gradient from deep navy (#0B1120) to burgundy (#3B0A1A) accented with luminous amber (#F59E0B) connection lines and cool slate typography. All on-screen elements, storyboard cards, timeline tracks, and captions strictly adhere to mobile platform safe margins: 120px top header clearance, 240px bottom UI clearance, and 80px side margins.

3. Multi-Agent System Architecture & Pipeline Data Flow

The orchestrator (core/orchestrator.py) manages a strictly typed directed acyclic graph (DAG) where each stage produces an immutable JSON contract in outputs/. The Critic agent continuously evaluates the entire artifact graph; if validation rules fail, a targeted repair loop re-executes only the earliest failing stage and its dependent downstream steps, capped at a maximum of three iterations.



4. Contract-Driven Orchestration & Bounded Repair Loop Mechanics

1. Stage Isolation & Immutable JSON Contracts: Every agent stage consumes strictly validated Pydantic models and emits persistent artifacts:
- Planner: Emits plan.json (enforcing duration window [35s, 45s], target 40s, 3-8 success criteria, and strict budget caps).
- Researcher (RAG): Grounded query engine for data/ (.pdf/.txt/.md), extracting facts.json with verified verbatim source quotations.
- Scriptwriter: Translates facts into script.json and script.md with speech-rate duration math (target ~180 wpm, bounds 120-190 wpm).
- Storyboarder: Designs 6-10 visual scenes in storyboard.json, enforcing on-screen text safe-length constraints (<= 80 chars per cue).
- Asset Generator: Synthesizes 1080x1920 RGBA frames and visual gradient cards cataloged in asset_manifest.json.
- Timeline Editor: Formulates edit_plan.json, aligns 48 kHz voiceover audio, burns SubRip captions, and muxes final MP4.
- Critic / QA: Audits 18 deterministic checks in qa_report.json. If issues are detected, the Orchestrator initiates an automatic repair pass.

2. Deterministic State Machine & Bounded Repair: When the Critic flags a validation failure (e.g. caption length overflow or duration mismatch), the orchestrator identifies the earliest responsible agent (e.g. Storyboarder for captions, Scriptwriter for speech duration) and restarts execution downstream from that exact point, feeding error diagnostics into the prompt. The loop is strictly capped at 3 passes to prevent runaway spend.

3. Local Offline Execution: Crucially, rendering and verification execute 100% locally on Apple Silicon without third-party cloud render dependencies.

5. Detailed Agent Roles, Implementation & Tooling

Agent	Core Responsibility	Input Dependencies	Output Contract	Primary Tooling
1. Planner	Decomposes brief into stages, constraints & budget limits	briefs/practice.yaml	outputs/plan.json	Pydantic v2, Groq LLM Adapter
2. Researcher	Grounded RAG fact extraction & verbatim quote verification	data/, plan.json	outputs/facts.json	PyMuPDF (fitz), Python regex
3. Scriptwriter	Crafts timed spoken narrative (Hook, Body, CTA) @ 180 wpm	plan.json, facts.json	outputs/script.json, script.md	Groq Adapter, duration math
4. Storyboarder	Plans 6-10 visual shots, layout prompts & on-screen text	script.json	outputs/storyboard.json	Groq Adapter, caption limiter
5. Asset Gen	Renders 9:16 gradient cards & procedural motion components	storyboard.json	outputs/asset_manifest.json	Pillow 13.0, vector math
6. Editor	NLE timeline assembly, 48kHz audio sync & SubRip captions	manifest.json, storyboard.json	outputs/edit_plan.json, MP4	FFmpeg 9.0.2, macOS say
7. Critic / QA	18 automated QA checks + hallucination & duration guards	All intermediate JSONs	outputs/qa_report.json	Deterministic validators, QA logic

6. Technology Stack & Local Rendering Architecture

• Core Foundations: Python 3.12 (macOS Apple Silicon). Data integrity across all seven agents is enforced via Pydantic v2 data models.
• Direct FFmpeg Byte Streaming: Cloud video generation APIs are non-deterministic black boxes that prevent automated verification. Instead, AgentReel combines Pillow (PIL 13.0) procedural graphics with FFmpeg 9.0.2. Crucially, the local Homebrew FFmpeg build lacked libass and libfreetype libraries, preventing standard subtitle filters (subtitles=... or drawtext=...). To overcome this platform limitation, agents/renderer_v3.py pipes procedural raw RGB24 bytes directly into FFmpeg stdin (-f rawvideo -pix_fmt rgb24 -s 1080x1920 -r 30 -i -). This guarantees 100% pixel-perfect typography, mobile safe-margin compliance, zero temporary image disk I/O, and ultra-fast rendering (>75 fps, >2.5x real-time speed).
• Audio Normalization: Local macOS speech synthesis (say -v Samantha) with automated audio padding (apad=whole_dur) and stereo silence fallback (anullsrc).

7. Development Progression: Milestone B through V3 Polish

Milestone / Version	Key Engineering Objectives & Deliverables	Verified Status & Preserved Evidence
Milestone B	Seven agents, orchestrator, repair loop, 63 unit tests. Live Groq LLM run (run_20261021_165923) generating initial plan, script, storyboard, and QA report.	outputs/plan.json -> qa_report.json; preserved in milestone_c_backup/
Milestone C	Card-based video rendering engine, macOS voiceover synthesis, SubRip captions, 39.0s vertical MP4 satisfying all Milestone C deliverables.	outputs/final_video.mp4 (3.06 MB), subtitles.srt; preserved in milestone_c_backup/
V2 Upgrade	Topic upgrade ('How 7 AI Agents Build a Video'), procedural motion graphics (5 animated scenes), raw RGB FFmpeg streaming, 43.02s vertical video.	outputs/final_video_v2.mp4 (43.02s, 2.74 MB), preview_v2_[1-5].png; preserved in v2_backup/
V3 Final Polish	Scene 2 audio audit & fix, positive audio headroom across all scenes (+0.29s to +0.99s), non-clipped storyboard cards, safe-margin compliance.	outputs/final_video_v3.mp4 (44.02s, 2.21 MB), outputs/AgentReel_Final.srt, 78 tests passing.

8. Automated Test Suite & Continuous Verification (78 Tests)

78 Unit & Integration Tests Passing in 2.66s • 100% Test Coverage Across Schema, Logic, Audio & Video Pipelines

Test Suite Module	Tests	Target Pipeline Component	Audited Invariants & Guardrails
tests/test_schemas.py	8 tests	Pydantic Schema Contracts	Duration [35s, 45s], target 40s, 3-8 criteria, 6-10 shots, valid stage enums
tests/test_validators.py	17 tests	Mathematical & Claim Validators	Speaking rate 120-190 wpm, gapless timeline continuity, ungrounded claim rejection
tests/test_critic.py	19 tests	Critic Rules & Safety Guardrails	Hallucinated number/contact detection, duration mismatch >1.0s, mandatory AI disclosure
tests/test_orchestrator.py	19 tests	Pipeline Orchestration & State	Targeted repair loops, earliest-failing rollback, max 3-pass termination safety
tests/test_renderer.py / motion	15 tests	Video & Subtitle Verification	1080x1920 canvas, SubRip sync (44.02s), Scene 2 seven-agent audit, backup integrity

Test Coverage Synthesis: The 78 automated tests provide continuous verification across all seven agent stages, schema contracts, arithmetic limits, and media output files. Testing execution time of 2.66s enables rapid local regression cycles prior to video synthesis. Dual backup verification confirms that earlier Milestone B and C deliverables (milestone_c_backup/) and V2 motion graphics assets (v2_backup/) remain untouched and independently reproducible.

9. Root-Cause Investigation: Scene 2 Audio Truncation & V3 Resolution

• Incident: In the V2 release, Scene 2 ('The 7 AI Agents Workflow') voiceover cut off abruptly, omitting the Editor and Critic agent roles before Scene 3 began.
• Mathematical Diagnosis: Spoken text required 19.25 seconds at natural speed (175 wpm). However, Scene 2 was allocated only 15.50s in agents/renderer_v2.py. During audio track concatenation, FFmpeg executed 'ffmpeg -i temp.aiff -af apad=whole_dur=15.5 -t 15.5 temp_padded.wav'. The '-t 15.5' hard-clipped the audio stream, dropping the final 3.75s of speech.
• V3 Engineering Resolution: 1) Refined narration to concise, natural phrasing explicitly naming all seven agents: 'In our pipeline, seven specialized agents coordinate: Planner sets the plan, Researcher checks facts, Scriptwriter writes the story, Storyboarder designs shots, Asset Generator creates visuals, Editor builds the edit, and Critic verifies quality.' (spoken duration: 15.90s at 180 wpm). 2) Adjusted Scene 2 duration to 16.50s, creating +0.60s of clean breathing headroom. 3) Audited all 5 scenes for positive headroom (+0.29s to +0.99s), ensuring zero audio truncation and exactly 44.02s total duration.

10. Runtime Economics, Token Audit & Latency Profile

Audited from logs/cost_report.csv and logs/run_log.jsonl (Run ID: run_20261021_165923, Model: openai/gpt-oss-120b on Groq). Zero metrics are simulated. Rendering for V3 executed 100% locally in 12.44s with 0 API calls.

Pipeline Stage	LLM Calls	Prompt Tokens	Completion Tokens	Total Tokens	Stage Latency
1. Planner	2	2,030	589	2,619	2.09s
2. Researcher (RAG)	0 (No local docs)	0	0	0	0.001s
3. Scriptwriter	2 (incl. repair)	3,749	1,715	5,464	25.56s
4. Storyboarder	2 (incl. repair)	3,708	1,504	5,212	27.29s
5. Asset Generator	0 (Procedural PIL)	0	0	0	0.119s
6. Editor	0 (Local assembly)	0	0	0	0.001s
7. Critic / QA	2 (incl. repair)	2,537	763	3,300	14.62s
TOTAL PIPELINE RUN	**8 calls**	**12,024 tokens**	**4,571 tokens**	**16,895 tokens**	**69.68s**

11. Human Oversight & Autonomy Analysis

Human review thresholds were configured in core/brief.py to halt execution if sensitive content (e.g. medical, financial, or political topics) is detected. For the educational brief P00 ('A day at the Agentic AI Lab'), requires_human_review evaluated to False. As audited in logs/interventions.md, exactly zero manual interventions were triggered throughout the entire generation process. The repair loop successfully resolved prompt schema inconsistencies and on-screen caption length limits deterministically without human assistance.

12. Limitations, Safety & Ethical Guardrails

• Limitations: 1) Local document corpus: In the live run, data/ contained no external PDF files; the Researcher honestly identified the absence of external sources, deriving facts strictly from brief statements without hallucinations. 2) Voice variety: TTS relied on macOS Samantha; multi-speaker dialogue requires local neural voice models. 3) Caption burning: Overcame missing FFmpeg libass/drawtext filters by streaming Pillow-rendered RGBA frames directly into rawvideo.
• Safety & Disclosure: Anti-hallucination validators forbid invented statistics, contacts, or ungrounded claims. Mandatory AI disclosure ('Created with AI agents — LY Agentic AI Lab') is permanently burned into the final scene's mobile-safe area.
• Reproducibility: Deterministic seed (seed=42), pinned dependencies in requirements.txt, and zero-network local rendering scripts (scripts/render_v3.py).
• Future Roadmap: 1) Local open-source LLMs (Llama 3.3 / Qwen 2.5 via Ollama) for 100% offline generation. 2) Local neural TTS (Piper/Coqui). 3) Apple Silicon VideoToolbox acceleration.

13. Final Deliverables Manifest & Submission Verification

Deliverable File	Technical Specifications / Details	Verification Status
outputs/final_video_v3.mp4	1080x1920 (9:16) MP4 | 30 fps | 44.02s | H.264/AAC | 2.21 MB	Verified playable, 100% clean decode
outputs/AgentReel_Final.srt	SubRip Subtitles | 6 cues | Exactly synchronized to 44.02s	Audio-verified, all 7 agents captioned
outputs/render_report_v3.json	JSON Manifest | Scene timings, codecs, bitrates, audio stats	Verified schema compliance
outputs/preview_v3_[1-5].png	1080x1920 PNG | 5 scene preview frame snapshots	Verified mobile safe-margin layout
reports/AgentReel_Architecture.mmd	Mermaid Diagram Source | Editable multi-agent flow & repair	Validated Mermaid syntax
reports/AgentReel_Process_Report.pdf	3-Page Engineering Report | Technical layout & audit	Generated via PyMuPDF (fitz)
outputs/v2_backup/ & milestone_c_backup/	Preserved earlier Milestone B, C, and V2 outputs intact	Verified dual backup integrity

Engineering Conclusion & Lab Sign-Off: The AgentReel challenge demonstrates that autonomous multi-agent pipelines can reliably produce broadcast-ready vertical video through modular decomposition, strict schema contracts, and bounded self-repair loops. With all 78 automated tests passing, 100% clean video decode, positive audio headroom across all scenes, and zero manual interventions, the codebase and deliverables stand fully verified and ready for formal submission at the LY Agentic AI Laboratory, K J Somaiya School of Engineering.

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